

SUPERMARKET SALES ANALYSIS

Interactive Power BI Dashboard Project

Project Presentation

Project Area	Data Analytics / Business Intelligence
Tool	Microsoft Power BI
Data Source	Excel
Dashboard Pages	6

1. Project Objective

The objective of this project is to develop an interactive Power BI dashboard that analyzes supermarket sales data and converts transaction-level information into meaningful business insights.

- Analyze overall sales, profit, orders, customers and profit margin.
- Understand sales trends and payment-method patterns.
- Compare product and category performance.
- Compare branch/store performance.
- Understand customer segments and purchasing patterns.
- Analyze profitability across categories and stores.

2. Business / Analytical Problem

A supermarket generates sales data across products, customers, stores, dates, payment methods and transactions. Without a consolidated analytical view, it can be difficult to identify sales patterns, product performance, branch differences, customer characteristics and profit contribution.

The dashboard provides a visual, data-driven way to explore these patterns and support business analysis.

Key questions addressed

- What are the overall sales, profit and order levels?
- How do sales change over time?
- Which products and categories generate more sales?
- Which stores contribute more sales and profit?
- How do customer types and demographic groups differ?
- Which categories and stores contribute to profitability?

3. Scope

In Scope

Area	Analysis
Sales	Total sales, sales trends, payment methods, customer type and quantity
Products	Product-level sales, quantity, categories and product details
Branches	Store sales, store profit, orders and branch details
Customers	Customer type, gender, age and customer-level summary
Profit	Profit by category, store, trend and sales-versus-profit comparison

Out of Scope

The project focuses on descriptive and comparative analysis of the supplied dataset; it does not claim that one variable causes another.

4. Data & Power BI Model

The project uses an Excel dataset containing transaction data and supporting dimension tables. The model is organized around the Sales table with related Product, Customer and Store tables.

Table	Purpose
Sales	Transactions, date, quantity, payment method, rating, sales, cost and profit
Product	Product name, category and unit price
Customer	Customer name, gender, age and customer type
Store	Store name, city and state

Core measures: Total Sales, Total Profit, Total Quantity, Total Orders, Total Customers, Average Order Value, Profit Margin, Total Cost and Average Rating.

5. Dashboard Pages

Dashboard 1: Executive Overview

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Profit Margin
- Sales by Category

Dashboard 2: Sales Analysis

- Monthly Sales Trend
- Sales by Payment Method
- Sales by Customer Type
- Quantity Sold by Category

Dashboard 3: Product Analysis

- Sales by Product
- Quantity Sold by Product
- Sales by Category
- Product Performance Details

Dashboard 4: Branch Analysis

- Sales by Store
- Profit by Store
- Orders by Store
- Branch Performance Details

Dashboard 5: Customer Analysis

- Sales by Customer Type
- Sales by Gender
- Customers by Age
- Customer Details

Dashboard 6: Profit Analysis

- Profit by Category
- Profit by Store
- Monthly Profit Trend
- Sales vs Profit by Category

6. Target Users

- College project evaluators and faculty
- Data Analytics students and trainers
- Business or store management
- Analysts interested in retail sales performance

7. Key Dashboard Outcomes

The completed report provides a single interactive environment for exploring sales, products, branches, customers and profitability. Users can move from high-level KPIs to detailed product, branch and customer views.

The report is designed for descriptive analysis and comparison and should be interpreted within the limits of the supplied dataset.

8. Tools & Technologies

Tool / Technology	Use
Microsoft Power BI	Data modeling, DAX measures, interactive dashboards and visualization
Microsoft Excel	Source dataset
DAX	Measures and analytical calculations

Thank You